CU-400 Condensing Unit: Service Manual

Document	Revision	Effective	Owner
NG-MAN-CU400	C	2025-03-01	Northgate Equipment Co.

1. Overview

The Northgate CU-400 is an air-cooled, semi-hermetic condensing unit for walk-in coolers from 800 to 2,400 cu ft. It runs on R-448A and holds box temperatures between 33F and 40F. The unit reports box temperature, head (discharge) pressure, power draw and cabinet vibration to the site monitoring system every 15 minutes.

2. Specifications

Item	Value
Refrigerant	R-448A, 14 lb factory charge
Compressor	Semi-hermetic reciprocating, 4 HP
Electrical	208-230V / 3 ph / 60 Hz, MCA 22 A, MOPD 35 A
Condenser	Copper tube, aluminum fin, 2 x 18 in fans, 1,075 rpm
Rated capacity	28,500 BTU/h at 25F SST, 95F ambient
Max ambient	115F
Weight	385 lb

3. Normal Operating Ranges

Readings outside these ranges for more than 30 minutes need attention, even if no fault code is active.

Measurement	Normal range	Alarm limit
Box temperature	34F to 38F	Above 41F for 30 min (E-42)
Head pressure	220 to 280 psig	Above 300 psig for 10 min (E-21)
Power draw	3.8 to 4.6 kW	Above 6.0 kW (E-30)
Cabinet vibration	Below 3.0 mm/s RMS	Above 5.0 mm/s

Fault Code Reference

Code	Name	Trigger condition	Severity
E-21	High head pressure	Head pressure above 300 psig for 10 consecutive minutes. Unit stops the compressor at 395 psig (high-pressure switch).	High
E-14	Low suction pressure	Suction pressure below 18 psig for 15 minutes.	Medium
E-30	Compressor overload trip	Power draw above 6.0 kW, or the internal overload opens.	High
E-42	High box temperature	Box temperature above 41F for 30 consecutive minutes.	High

Troubleshooting by Fault Code

E-21: High head pressure

Trigger: Head pressure above 300 psig for 10 consecutive minutes. Unit stops the compressor at 395 psig (high-pressure switch).

Likely causes, most common first:

- Dirty or blocked condenser coil. This causes about 70% of E-21 events, especially at hot or dusty sites.
- Failed condenser fan motor or seized fan blade.
- Recirculation of hot discharge air because of blocked clearance (needs 24 in on all sides).
- Refrigerant overcharge after recent service.
- Non-condensable gas in the system.

Corrective actions:

1. Check that both condenser fans run. A fan that is off with the compressor running points to a failed fan motor or capacitor.
2. Inspect the condenser coil. If dirty, apply LOTO and clean with coil cleaner and low-pressure water from the inside out.
3. Confirm 24 in clearance around the unit and remove debris, pallets or shrink wrap.
4. If pressure stays above 300 psig after cleaning, compare subcooling to the 10F to 14F target. High subcooling means overcharge.
5. Log readings before and after in the work order. If the box temperature has risen above 41F, start SOP-OPS-002.

Rising box temperature together with E-21 means the unit is losing capacity. Treat it as a P1 event under SOP-OPS-003.

E-14: Low suction pressure

Trigger: Suction pressure below 18 psig for 15 minutes.

Likely causes, most common first:

- Iced evaporator coil.
- Low refrigerant charge or leak.

- Restricted filter drier.

Corrective actions:

1. Check the evaporator for ice and run a manual defrost.
2. Leak test with an electronic detector.
3. Measure pressure drop across the filter drier. Replace if above 3 psi.

E-30: Compressor overload trip

Trigger: Power draw above 6.0 kW, or the internal overload opens.

Likely causes, most common first:

- High head pressure (see E-21).
- Low supply voltage (below 197V).
- Failing compressor valves.

Corrective actions:

1. Measure supply voltage on all three legs. Imbalance above 2% needs an electrician.
2. Clear any E-21 condition first, then reset.
3. If E-30 repeats within 24 hours, open a P2 work order for compressor diagnosis.

E-42: High box temperature

Trigger: Box temperature above 41F for 30 consecutive minutes.

Likely causes, most common first:

- Door left open or damaged door gasket.
- Any active E-21, E-14 or E-30 fault.
- Product loaded warm.

Corrective actions:

1. Check the door and strip curtain.
2. Resolve any other active fault code.
3. Start SOP-OPS-002 Cold Chain Temperature Excursion Response.

Preventive Maintenance

Task	Interval
Clean condenser coil	Quarterly. Monthly at Lakeland and Stockton sites (see SOP-OPS-004).
Check fan motors and blades	Quarterly
Check refrigerant charge and subcooling	Semi-annually
Tighten electrical connections	Annually

WARNING: Apply lockout/tagout per SOP-OPS-001 before opening any electrical enclosure or touching moving parts. Only authorized technicians may perform these steps.